



The perceptions and practices of parents and children on acute pain management among hospitalized children in two Botswana referral hospitals



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ABSTRACT

Background: Perceptions and practices of parties in pediatric pain are critical in children's access to adequate acute pain management. The personal factors of the child and parents have been shown to be central to pediatric pain management by the Symptom Management Theory.

Aim: To describe children and parents/guardians' perceptions (knowledge, attitudes and beliefs) and practices regarding pediatric acute pain management and explain the influence of socio-cultural and environmental factors on those perceptions and practices.

Methods: Descriptive cross-sectional survey using modified versions of the American Pain Society Patient Outcome Questionnaire-Revised among parents/guardians and children.

Results: A convenience sample of 275 parents/guardians and 42 children aged 8 to 13 years admitted between date November 2018 and February 2019 to two Botswana tertiary hospitals completed the surveys. Forty-seven percent ($n = 129$) of parents/guardians reported the child to be in moderate-severe pain, while 38% ($n = 16$) of children reported pain as moderate-severe at the time of the survey. The children mean scores for cm-APS-POQ-R were 113(33) while parents/guardian's guardians for m-APS-POQ-R were 123(26). The subscales except for the parents/guardians' pain interference ($p = .96$) were statistically significant ($p = .000$), showing adequate knowledge, positive attitudes and high pain intensity for both parents/guardians and children.

Conclusion: Parent/guardians and children reported a high incidence of acute pain, were content with pain management services, and showed adequate knowledge of pediatric pain and its management. The incongruence between the intensity of pain, satisfaction on the adequacy of pain management and knowledge and attitudes demonstrated in this study need further inquiry.

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Background

Pain is a common symptom among individuals with various diagnoses that, if ignored, can have detrimental effects on the health outcomes, perceptions of the quality of care and subsequent quality of life (Dancel

et al., 2017; Schellack & Matimela, 2016). Detrimental effects of pain include short-term and long-term effects (Dancel et al., 2017; Henschke et al., 2015; Schellack & Matimela, 2016; Schwaller & Fitzgerald, 2014). Short-term effects result from neuro-hormonal responses such as hemodynamic instability, metabolic alterations, and impaired immune response, as well as suffering (Dunwoody et al., 2008; Low & Schweinhardt, 2012; Schwaller & Fitzgerald, 2014), while long-term effects include permanent central and peripheral neurologic changes-peripheral sensitization and central sensitization and psychological impact (Dunwoody et al., 2008; Low & Schweinhardt, 2012; Schwaller &

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Fitzgerald, 2014). Unfortunately, despite the significance of pain in routine care, literature suggests significant numbers of children admitted to health care facilities often suffer from inadequately managed pain (Angelini et al., 2018; Azam et al., 2012; Dancel et al., 2017). While this is a global problem, children in Low- and Middle-Income Countries (LMIC) are at even higher risk of experiencing inadequately managed pain due to environmental, socio-cultural, developmental factors, and the subjective nature of pain (Alotaibi et al., 2018; Bond, 2011; Finley et al., 2009; Matula et al., 2018). Environmental factors in LMIC relate to resource limitations including drug formulations for children, unreliable supply of drugs, and lack of adequately trained health care providers (Albertyn et al., 2009; Clancy, 2014). Socio-culturally, children in LMIC are considered not to feel pain as adults do and to be stoic, and parents and healthcare providers often do not believe children's pain self-reports (Batista et al., 2012; Bosenberg, 2007; Finley et al., 2009; Forgeron et al., 2009). The socio-cultural milieu arises from the lack of knowledge, beliefs, attitudes towards management and practices related to pediatric acute pain (Finley et al., 2009; Matula et al., 2018).

Pediatric acute pain is complex owing to developmental stages of children and subjective nature of pain, therefore, requires a well-versed caregiver for its recognition and management. The recognition and management of pediatric pain requires a series of engagement between involved parties being the child, parents/guardians, and healthcare providers (Matula, 2019). This engagement is further illustrated in a model derived from the Symptom Management Theory which is used to guide this study (Fig. 1) (Matula, 2019). The Symptom Management Theory suggests that individual's perception, evaluation of meaning and response to symptom are influenced by personal, family, health and illness, and environmental related factors (Humphreys et al., 2008). Therefore, pediatric pain is influenced by factors related to health, environment and the social system (Angelini et al., 2018; Matula et al., 2018). Hence, a lack of knowledge by any involved party may lead to impediments to child pain management (Forgeron et al., 2009; Matula et al., 2018). Evidence has shown that lack of knowledge

by parents/guardians is a significant barrier towards pediatric pain relief (Olaogun et al., 2008). Evidence also reports a conflict between parents/guardians in LMIC cultural beliefs and those of the healthcare providers due to their challenges related to their overall understanding of pain (He et al., 2006; Jongudomkarn et al., 2012). Therefore, knowledge and understanding of children and their parents/guardians related to pain and its management must be considered as this is essential in identifying areas that need strengthening to improve pain management in this population.

Pain is subjective and requires to be recognized and communicated to health care providers for management, therefore, this requires the parent/caregiver's and child's astute and positive attitudes towards recognition of child pain and the associated need for it to be managed (Aziznejadroshan et al., 2015; Chng et al., 2015; Kristjansdottir et al., 2018; Schellack & Matimela, 2016). Therefore, the individual parties in child pain idiosyncrasies embedded in socio-cultural and environmental contexts often influence how the child and parents/guardians respond to pain and request services related to pediatric pain management (Kristjansdottir et al., 2018; Pope et al., 2017). Evidence reports that children and parents/guardians who reside in LMIC, such as Botswana, face unique situations such as high burden of childhood diseases, limited resources, inadequately trained health care providers and culture that impede child pain treatment which informs their perceptions, decisions, and experiences related to pain care (Matula et al., 2018). The respect “*Botho or Ubuntu*” culture further make child pain treatment intricate, with parents/guardians and children often not requesting or communicating the pain needs due to respect. For instance, Forgeron et al. (2009) suggest that health care providers' viewed as superiors or knowledgeable by parents/guardians and children in LMIC impacts their ability to seek pain relief as parents/guardian and children afford them respect by not asking for pain relief.

This study studies perceptions and practices of children and parents/guardians, which encompasses personal, family and health and illness related factors. Documenting perceptions of children and

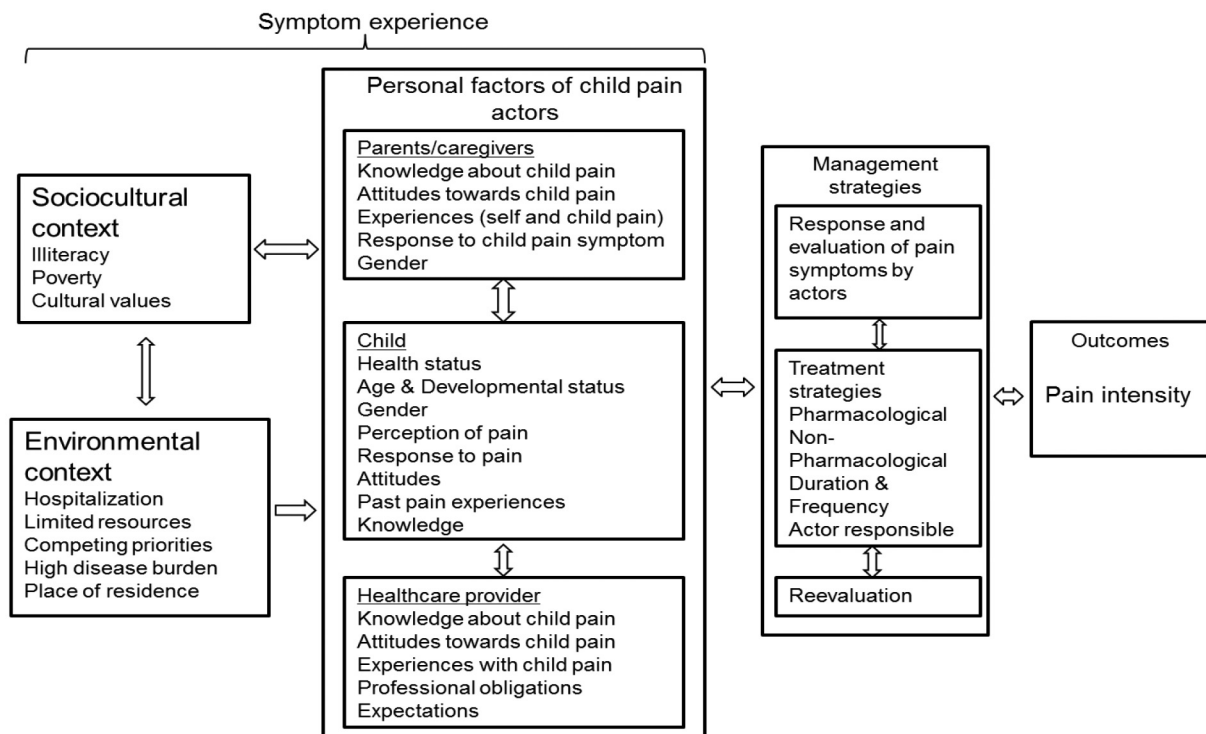


Fig. 1. Conceptual Model: Pediatric pain symptom management derived from Symptom Management Theory (Humphreys J, Lee KA, Carrieri-Kohlman V, Puntillo K, Faucett J, Janson S, et al. 2008, p. 145–58).

parents/guardians provides critical patient-reported information to reflect on the understanding and expectation of parents/caregivers and children and identify potential socio-cultural barriers that impede pediatric acute pain management. Therefore, the purpose of this study is to describe children and parents/guardians' perceptions-knowledge, attitudes and beliefs and practices regarding pediatric acute pain management. Furthermore, this study will explain the influence of socio-cultural and medical factors on the children and parents/guardians' perceptions and practices regarding pain management in two Botswana tertiary hospitals.

Methods

Study design

A descriptive cross-sectional survey was conducted, as it is the most relevant design for assessing knowledge, attitudes, and patient-reported outcomes (Kesmodel, 2018).

Population and sample

The study population is parent/caregiver-child dyads admitted to the two referral hospitals in Botswana. Based on Botswana health guidelines, a pediatric patient is aged between 0 months to 13 years. Using convenience sampling, the study was conducted in the pediatric units of two tertiary referral hospitals in Botswana, Nyagabgwe Referral Hospital (NRH) and Princess Marina Hospital (PMH). These two public hospitals serve as important pediatric centers across the country with a diverse population and an average of 6000 children admitted annually between the two hospitals.

Participant eligibility

Participants were included in the study if they were: a) parent/caregiver-child dyad with the child being aged two months to 13 years admitted to a pediatric unit; b) fluency in spoken English or Setswana and c) for, children ≥ 8 years, if they could complete a survey. Participants were excluded if: a) children were admitted in the neonatal ward or were < 2 -months of age; b) children were admitted solely for nutritional rehabilitation, non-medical grounds, such as awaiting transfer to other hospitals, abandoned children, and forensic cases; c) child or parent with apparent developmental delay or cognitive impairments; d) parent not present or e) child was admitted for observation for less than 24 h.

Variables and measures

Demographic variables for children

Demographic variables were collected through self-report and health records; this included age, gender, admitting diagnosis, surgical procedure, and admission unit.

Demographic variables for parents/guardians

Demographic variables were collected through self-report, variables include age, gender, relationship to the child, residential location, educational level, and parents'/guardians' perception of the child's illness acuity.

Perceptions and practices of parents/guardians

Parents/caregivers perspectives and practices were collected using the Modified American Pain Society Patient Outcome Questionnaire-Revised (m-APS-POQ-R) (Gordon et al., 2010). The APS-POQ-R is a cross-cultural instrument validated for hospitalized adults (Ali et al., 2016; Botti et al., 2015). This tool, the APS-POQ-R, has also been validated as a proxy measure for parents in pediatric pain and used for caregiver's perspectives in some high-income countries (HIC) (Cronbach's

alpha of 0.798) (Chaw et al., 2019). The m-APS-POQ-R is a 28-scale tool with four domains of pain severity, perception of care, pain interference, and knowledge derived from a scale used by Ali et al. (2016). An overall score ≥ 93 and score ≥ 20 for knowledge subscale, score ≥ 26 for perception subscale, score ≥ 30 for pain interference subscale and a mean pain severity subscale score of ≤ 24 indicate better pain outcomes. Questions related to pain severity (1–4), pain interference (5 and 6), impact on parent mood (10), and the importance of pain treatment (8) were reversed for analysis. In addition, all knowledge questions (11) except the last question were reversed for analysis.

Child perceptions and practices

In Botswana, sometimes children who are eight (8) years and above can usually self-report pain and are often admitted without guardians, thus being responsible for reporting their symptoms. Their inclusion provides valuable information from a child's perspective (He et al., 2007; Pope et al., 2017). The Child Modified version of APS-POQ-R (cm-APS-POQ-R) was developed from the m-APS-POQ-R to capture children's perceptions and practices. The cm-APS-POQ-R is a 28-scale tool with four domains of pain severity, perception of care, pain interference, and knowledge derived from a scale used by Ali et al. (2016). An overall score ≥ 93 and score ≥ 20 for knowledge subscale, score ≥ 26 for perception subscale, score ≥ 30 for pain interference subscale and a mean pain severity subscale score of ≤ 24 indicate better pain outcomes. Questions related to pain severity (1–4), pain interference (5 and 6), impact on parent mood (10), and the importance of pain treatment (8) were reversed for analysis. In addition, all knowledge questions (11) except the last question were reversed for analysis. Notably the cm-APS-POQ-R uses language at reading level of an 8–10-year-old for easy understanding by children. The cm-APS-POQ-R has not been validated in children, however it is derived from a validated tool APS-POQ-R. The advantage of using the same conceptual instrument for both children and parents/guardians is to ensure that the study results are comparable.

Pain intensity

The Numerical Rating Scale (NRS) was used for the parent/guardian's.

assessment of children's pain. The NRS has been widely used for pain measurement in adults. As a proxy measure for children's pain, the NRS has shown consistency in various populations (Huang et al., 2012; Khin Hla et al., 2014). The c-statistic in this sample was 0.80 (Matula et al., 2022). The Faces Pain Scale-Revised (FPS-R) is a standard self-report pain measurement tool used with pediatric patients. The FPS-R consists of six gender-neutral faces depicting a face of "no pain expression" on the left (scored as 0) and a face of "most possible pain" on the right (scored as 10) (Hicks et al., 2001). It is validated in this population with a c-statistic of 0.70 (Matula et al., 2022).

Bias

The researchers were aware of the likely response bias to investigators due to perception of authority and fear of appearing ungrateful and how the complaints might affect future care. To address this bias, all the researchers were not part of the treatment team and it was further explained during the consent that the study did not influence the care and the researchers only collected data when the parents/guardians did not have any treatment team present. The m-APS-POQ and cm-APS-POQ were not validated in this population but has strong cross-cultural psychometrics and the internal consistency of $\alpha = 0.84$ and 0.80 respectively.

Sample size

The sample size was estimated based on statistical methods for descriptive data analysis using Daniel (1999) method (Eq. 1). According to the equation;

$$N = 4(Z_{crit})^2 p(1-p) \div D^2 \quad (1)$$

where N is the sample size, $Z = 1.96$ (95% CI), $D = 0.22$ ($1-p$) and $p = .78$ - the proportion of children experiencing pain as population estimate from previous study leading to a sample size of 55 (van der Heijden et al., 2018). Considering multisite, subgroup, and correlation analysis for some variables, the sample size is quadrupled to 220. The attrition rate set at 20%, resulting in an estimated sample size of 264, was required for this study.

Ethical considerations

The study was approved by the University of Pennsylvania (830264) and the University of Botswana (UBR/RES/IRB/BIO/105) Institutional Review Boards (IRB), the Botswana Ministry of Health and Wellness Health Research and Development Committee (HRDC) (REF: HPDME 13/18/1), and the ethics review committees of the respective institutions (REF: NRH 1/2/170 and REF: PMH5/79 (464–12,018)) respectively. Self-reports are considered to be the gold standard for reporting pain (Hicks et al., 2001), therefore the children reports were considered to be a true reflection of the child pain experiences. To reduce bias from power in relation of the child, parent/guardian and staff, the researcher were not members of the staff and did not wear clothing associated with members of the treatment staff. Furthermore, the participants were reassured about the confidentiality of their data and how the data is going to be used to allay anxiety related to retributions of complaining. While data was collected in the unit, the researchers picked the time when the parent/guardians were free from treatment team to reduce bias of being overhead.

Procedures

The study was conducted over 11 weeks between November 2018 and February 2019, except on holidays in both English and Setswana. Data collection was conducted by the principal researcher and two research assistants fluent in both English and Setswana. Consent and Assent were obtained by the principal researcher. Participants were screened for inclusion and approached about the study after 12 h of hospital admission; following enrollment, the survey was conducted between 48 and 60 h of stay in the hospital. Study data were collected and managed using the Research Electronic Data Capture tool (REDCap) (Harris et al., 2009; Harris et al., 2019;) electronic data capture hosted at the University of Pennsylvania. REDCap is a secure, web-based software platform designed to support data capture for research studies, providing 1) an intuitive interface for validated data capture; 2) audit trails for tracking data manipulation and export procedures; 3) automated export procedures for seamless data downloads to standard statistical packages, and 4) procedures for data integration and interoperability with external sources (Harris et al., 2009; Harris et al., 2019). All analyses were conducted in Stata version 15.1 (State Corp, LP, College Station, TX, USA). The significance level for statistical tests is set at ≤ 0.05 . Participants with missing data were treated as 0 during analysis.

Statistical analysis

Descriptive statistics, including means, standard deviations (SD), ranges, proportions, and percentages, were used for sample baseline characteristics and variables of interest. A single sample t -test was used to compare the means of the outcome variables. Means for m-APS-POQ-R, cm-APS-POQ-R, knowledge, perception of care, and pain interference subscales were set at the population to mean 50%, assuming a normal distribution. The pain severity subscale mean was set at ≤ 24 , representing moderate-severe pain, which is often considered clinically relevant due to the need to be treated. ANOVA was used to analyze variance between the means of the outcome variables and some

characteristics of children and parents/guardians. P -value was set at 0.05 for all analyses.

Results

Participants characteristics

A total of 518 child-parent dyads were approached between November 21, 2018, and February 8, 2019; a total of 387 dyads agreed to participate (75% response rate). Two hundred and seventy-five (275) parents/guardians and 42 children ≥ 8 years of age completed the surveys. The mean age (SD) for parents/guardians was 32 years (10.2), ranging from 17 to 81 years, and for children was 10 years (1.6), ranging from 8 to 13 years (Table 1).

Pain intensity

Forty-seven percent ($n = 129$) of parents/guardians reported that their children were experiencing moderate-severe pain, while 37% ($n = 102$) indicated their children did not have pain at the time of the survey. Fifty-nine percent ($n = 162$) and 52% ($n = 143$) of parents/guardians reported moderate-severe pain as the child's worst and average pain, respectively. Thirty-one percent ($n = 107$) and 29% ($n = 80$) reported no pain as average and worst pain the child had, respectively. Thirty-eight percent ($n = 16$) of children reported being in moderate-severe pain, while 45% ($n = 19$) stated that they were not in pain. The average pain was 57% ($n = 24$) for moderate-severe pain and 36% ($n = 15$) for no pain, while the worst pain was reported at 60% ($n = 25$) for moderate-severe pain and 35% ($n = 15$) for no pain.

Pain perceptions and practices and subscales scores

The parent's and children's mean scores for m-APS-POQ-R, $p = .000$ and cm-APS-POQ-R, $p = .000$ were statistically significant when compared to the population means (Table 2). In addition, the means of the following m-APS-POQ-R subscales were statistically significant: pain severity, $p = .000$, perception of care, $p = .000$, and knowledge, $p = .000$; however, there was no statistically significance in pain interference, $p = .96$ compared to population means. All subscales were statistically significant, $p = .000$ in the children's survey (Table 2). The majority of parents/guardians, 59%, stated that a combination of pharmacological and non-pharmacological measures was used to manage children's pain. In comparison, 8% and 33%, respectively, indicated that pain was managed by either pharmacological or non-pharmacological measures. Also, a more significant number of children, 72%, indicated that a combination of pharmacological and non-pharmacological measures was mainly used to manage their pain. Nevertheless, only 8% and 21% indicated that pain was managed independently by either pharmacological or non-pharmacological measures.

Correlation of pain perceptions and practices scores and participants demographics

Parents/guardians aged ≤ 24 years were more likely to score higher on m-APS-POQ-R, $p = .04$, knowledge, $p = .015$, and pain severity, $p = .000$, subscales than other parents/guardians (Table 3). Parents/guardians of children admitted in medical units likewise had higher scores on m-APS-POQ-R, $p = .000$, and knowledge, $p = .000$, pain severity $p = .02$ perception of care, $p = .003$, subscale (Table 3). Parents/guardians in NRH surgical unit reported higher scores on pain interference, $p = .008$ (Table 3). Child diagnosis was similarly statistical significant in the scoring m-APS-POQ-R, $p = .019$, and the pain severity subscale, $p = .003$. The parents/guardians' perceptions of the child's illness acuity were significant in scoring m-APS-POQ-R, $p = .000$, pain severity, $p = .000$, perception of care $p = .000$, and pain interference, $p = .000$ (Table 3). Child diagnosis was the main determining factor

Table 1
Participant demographics and characteristics.

Characteristics	Mean(sd)	Range	Count(n)	Percent
Parents/guardians demographics				
Age (years)	32(10)	17–81	275	
Age groups (years)				
17 < 25			63	22.9
25 < 35			119	43.3
35 < 50			79	28.7
≥50			14	5.1
Gender				
Female			262	95.3
Relationship to child				
Mother			229	83.3
Grandmother			15	5.8
Father			13	4.7
Other			18	6.6
Education level ^a				
Elementary			135	49.1
High School			61	22.2
College			54	19.6
Degree and above			25	9.1
Residence				
Rural			118	42.9
Semi-Urban			58	21.1
Urban/town			33	12
City			66	24
Children's demographics				
Age (Years)	4(3.3)	1–13	275	
Age group (Years)				
<6			200	72.7
6–9			52	18.9
10–13			23	8.4
Gender				
Female			106	39.2
Hospital and Unit of Admission				
PMH-Medical			58	21.1
PMH-Surgical			75	27.3
NRH-Medical			82	29.8
NRH-Surgical			60	21.8
Common diagnosis group ^b				
Acute Gastroenteritis			30	10.9
Trauma			55	20
Pneumonia			22	8
Wounds			20	7.3
Cancers			13	4.7
Other			134	49.1
Length of child's Admission				
0 > 3 days			138	50.6
3 > 5 days			62	22.7
6 > 11 days			38	13.9
11 > 14 days			20	7.3
≥ 14 days			15	5.5
Child (< 8 years) respondent demographics				
Age (years)	10(2)	8–13	42	
Age groups				
8 < 10			21	50
10–13			21	50
Gender				
Male			27	64.3
Common diagnosis group ^c				
Cancer			6	14.7
Trauma			16	39
Other			19	46.3
Residence				
Rural			5	12.5

Table 1 (continued)

Characteristics	Mean(sd)	Range	Count(n)	Percent
Sub-urban			6	15
Urban/town			10	25
Cities			19	47.5
Unit Type				
Medical			7	16.7
Surgical			35	83.3
Hospital of admission				
PMH			25	59.5
NRH			17	40.5
Length of Child's Admission				
0–2 days			16	40
3–5 days			10	25
6–10 days			7	17.5
>10 days			7	17.5

Note: ^aeducation categories: elementary (1st to 10th grade), high school (grade 11 and 12), college (certificate and diploma holders) and degree and above for a university degree and above. ^bSix diagnostic groups are deduced from aggregating all related medical diagnoses: (1) Pneumonia, (2) Acute gastroenteritis, (3) Wounds –all open wounds that require dressing changes except motor vehicle, fractures and trauma-related injuries which are grouped as (4) Trauma, (5) Cancer, and all the remaining diagnoses were grouped as (6) Others. ^cDiagnostic groups are further consolidated for children survey by combining wounds and trauma to make 1) Trauma, 2) cancer and 3) Others which include pneumonia and acute gastroenteritis. PMH = Princess Marina Hospital, NRH = Nyangabgwe Referral Hospital.

on overall child scores on the cm-APS-POQ-R, $p = .04$ and perception of care score, $p = .03$ subscale scores (Table 4). The child age group was significant for the knowledge, $p = .007$ subscale, with children 8–9 years of age having lower scores. The child's place of residence was significant for pain interference, $p = .025$, subscale with children from less-resourced areas (rural/semi-urban) scoring higher than those from cities and towns (Table 4).

Discussion

This is the first study to quantitatively evaluate pain, perceptions and practices in parent/guardian-child dyads in Botswana's two referral hospitals. This is one of the few studies to consider children's understanding of pain and its management in sub-Saharan Africa, where children are thought to be stoic (Bosenberg, 2007; Jongudomkarn et al., 2012; Pope et al., 2017). Our findings are similar to those reported by Eshete et al. (2019), who found that patient satisfaction with pain care in sub-Saharan Africa was not correlated with pain intensity (Eshete et al., 2019). The survey results will inform further research exploring the adaptation of pain measures, management strategies and

Table 2
Overall pain care outcomes and subscales means.

	Scales and subscales	Mean (sd)	Range	p-value	r
Parent/guardian	m-APS-POQ-R	113.48 (33.22)	22–174	0.000	0.84
	Knowledge	25.78 (7.34)	0–40	0.000	0.59
	Perception of care	35.62 (11.03)	0–51	0.000	0.75
	Pain interference	31.88 (18.04)	0–54	0.96	0.87
	Pain severity	20.16 (8.5)	3–34	0.000	0.86
Child	cm-APS-POQ-R	122.79 (25.77)	58–171	0.000	0.80
	Knowledge	24.24 (7.07)	0–37	0.000	0.55
	Perception of care	37.76 (8.24)	6–51	0.000	0.77
	Pain interference	40.57 (16.03)	9–59	0.000	0.85
	Pain severity	20.21 (9.52)	0–31	0.000	0.89

Note: m-APS-POQ-R-Modified American Pain Society Patient Outcome Questionnaire-Revised-m-APS-POQ-R, Child Modified American Pain Society Patient Outcome Questionnaire-Revised-cm-APS-POQ-R, r = Cronbach's alpha, $p < .05$, maximum scores: m-APS-POQ-R = 187, Knowledge = 40, Perception of care = 52, Pain interference = 60, Pain severity = 35, and cm-APS-POQ-R = 187.

Table 3
Relationship between parent/guardian demographics and outcomes survey and subscale scores.

Characteristics	m-APS-POQ-R	Knowledge	Perception of care	Pain interference	Pain Severity
Parents age group	***	***	X	X	***
Child age group	X	***	X	X	***
Parent Gender	X	***	X	***	X
Child gender	X	X	X	X	X
Relationship to child	X	***	X	X	X
Hospital and Unit of Admission	***	***	***	***	***
Education level ^a	X	X	X	X	X
Common diagnosis group ^b	***	X	X	X	***
Residence	***	X	***	***	***
Child condition by parent	X	X	X	***	X
Length of Stay	X	***	X	X	X

Note: *** statistically significant = $p \leq .05$, X = not statistically significant. ^aeducation categories: elementary (1st to 10th grade), high school (grade 11 and 12), college (certificate and diploma holders) and degree and above for a university degree and above. ^bSix diagnostic groups are deduced from aggregating all related medical diagnoses: (1) Pneumonia, (2) Acute gastroenteritis, (3) Wounds –all open wounds that require dressing changes except motor vehicle, fractures and trauma-related injuries which are grouped as (4) Trauma, (5) Cancer, and all the remaining diagnoses were grouped as (6) Others.

interventions to improve patient satisfaction in Botswana and other countries in the region.

The results indicate a high burden of pediatric acute pain (38%–60%) among hospitalized children in Botswana, particularly moderate-severe pain, but children and parents/guardians, despite recognizing that pain is 'detrimental' to healing and requires treatment are content with the pain services provided. This calls for further investigation to deduce how children and parents/guardians' expectations and access to resources influence the expectations of pain treatment. The results of this regarding high pain burden are also supported by [Doca et al. \(2017\)](#) who found similar trends in pediatric acute pain prevalence in Brazil ([Doca et al., 2017](#)). However, their sample indicated that pain was not adequately managed, which is the opposite of the results of this study ([Doca et al., 2017](#)). Furthermore, the results of this study are comparable to the pediatric acute pain prevalence findings reported by [Harrison et al. \(2014\)](#) and [Walther-Larsen et al. \(2017\)](#), except that the majority in these samples reported mild pain. Regionally, the results are comparable to pediatric acute pain in pediatric sub-populations ([Azam et al., 2012](#); [van der Heijden et al., 2018](#)).

Unexpectedly results indicated that both parent/guardians and children have adequate knowledge about their pain. The knowledge subscale was statistically significant for both groups. Similar results were reported for parents' knowledge by [Chng et al. \(2015\)](#) and [Zhu et al. \(2018\)](#). Other authors have also reported that children have adequate knowledge about their pain ([Kortesluoma et al., 2008](#); [Twycross & Finley, 2013](#)). However, these results were not expected because most studies cite low levels of knowledge among children and parents/guardians because of illiteracy and cultural norms ([Matula et al., 2018](#)). The inconsistency between pain prevalence and service views by both parents/guardians and children may be best explained by inadequate knowledge.

The pain interference subscale was only statistically significant for children; these results were surprising. Parents/guardians did not consider pain interfering with the children's functioning despite the high pain intensity. Evidence suggests that parents/guardians believe acute

pain interferes with the child's activity of daily living ([Lim et al., 2012](#)), which is contrary to the findings of this study. More inquiry is needed on why parents/guardians did not consider pain interfering with a child's hospitalization and activities. This could be key to understanding how cultural norms and expectations may be influencing these parents and caregivers.

Parents scoring on the m-APS-POQ-R and the subscales was influenced by parental age, unit of admission, child diagnosis, and the perceived child illness acuity. In contrast, child diagnosis was the main factor in a child scoring the m-APS-POQ-R and subscales. Thus, the results suggest a demographic shift in understanding pain and illness associated with an apparent injury with pain interference. Also, the results suggest that children in the medical unit are less likely to be considered in severe pain when compared to those in the surgical unit or with illness with apparent injuries, which is a prevalent misconception globally ([O'Neal & Olds, 2016](#)).

Overall, this study adds new knowledge regarding the burden of pediatric pain and children and parents/guardians' view of pediatric pain management in two of Botswana's hospitals. The results of this study provide a baseline for further inquiry into children and parents/guardians views and understanding of the need to treat pediatric acute pain. The incongruity between pain assessment and the need for treatment must be explored further, along with non-pharmacological measures parents/guardians and children employ to lessen their pain. In addition, cultural influence on pain reporting and expectation for treatment need to be explored.

The limitations of this study include selection bias due to convenience sampling. Two referral hospitals were included; each admits a diverse population of children with various levels of illness acuity and demographics as patients and families are from rural and urban northern and southern Botswana, therefore bringing heterogeneity to the sample. Also, the scales used have not been validated in this population, and comparison groups or cut-off scores do not exist for these measures. The data were normally distributed (Shapiro and Wilk test statistic range of 0.88–0.98), however, the population means of 50% for the

Table 4
Relationship between children demographic and outcome survey and subscale scores.

Characteristics	cm-APS-POQ-R	Knowledge	Perception of care	Pain interference	Pain Severity
Age (years)	X	***	X	X	X
Gender	X	X	X	X	X
Common diagnosis group ^a	X	X	***	***	X
Residence	X	X	X	***	X
Unit Type	X	X	X	X	X
Hospital	X	X	X	X	X
Length of Stay (days)	X	X	X	X	X

Note: *** statistically significant = $p \leq .05$, X not statistically significant ^aDiagnostic groups are further consolidated for children survey by combining wounds and trauma to make 1) Trauma, 2) cancer and 3) Others which include pneumonia and acute gastroenteritis.

scale and all subscales except pain severity ≤ 24 partly address this limitation. Also the effect of the power differences often portrayed by parents/guardians and children in this cultures may have influenced the results. Finally, the generalizability of the results may be limited mainly due to cultural differences.

Practice implications

Children in Botswana may experience high pain intensity and are unlikely to seek and receive pain relief due to cultural respect for health care workers. Despite this, Parents and children in Botswana have adequate knowledge and positive attitudes towards pain management but their expectations may not match pain management needs. The modified versions of the American Pain Society Patient Outcome Questionnaire-Revised (APS-POQ-R) can be used as a tool to assess both parent/guardians and children's perceptions and practices across cultural settings. Lastly it should be noted that parents/guardians may not consider pain to be interfering with child's activities as they expect the child to be in bed when sick as opposed to be up and about or playing as usual.

Conclusion

Parents/guardians and children reports about pain treatment in Botswana referral hospitals are seemingly contradictory. That is, while they demonstrate adequate knowledge about pediatric pain and its management, believe children's pain should be treated, and are receptive to pain management, they are satisfied with pain management services despite high incidence of moderate-severe pain. This may be explained by the cultural belief that pain does not interfere with the recovery of children from illness requiring hospitalization. In addition, parents/guardians age, perceptions of child illness acuity and an apparent wound influenced their scores. In contrast, child age, apparent wounds and place of residence determined how the child scored the survey and their understanding and views on pain management.

CRediT authorship contribution statement

Samuel T. Matula: Conceptualization, Data curation, Formal analysis, Funding acquisition, Investigation, Methodology, Validation, Visualization, Writing – original draft, Writing – review & editing. **Sharon Y. Irving:** Conceptualization, Formal analysis, Investigation, Methodology, Supervision, Validation, Visualization, Writing – review & editing. **Janet A. Deatrick:** Conceptualization, Formal analysis, Investigation, Methodology, Supervision, Validation, Visualization, Writing – review & editing. **Andrew P. Steenhoff:** Conceptualization, Formal analysis, Investigation, Methodology, Supervision, Validation, Visualization, Writing – review & editing. **Rosemary C. Polomano:** Conceptualization, Formal analysis, Investigation, Methodology, Validation, Visualization, Writing – review & editing.

Declaration of Competing Interest

All the Authors have no conflicts of interest to disclose.

We confirm that this work is original and has not been published elsewhere, nor is it currently under consideration for publication elsewhere.

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